

ET AI Hackathon 2026

Problem Statement 6: AI for Digital Public Safety

Citizen Fraud Shield – An AI-Powered Platform for Detecting Digital Fraud and Scam Activities

Executive Summary

Citizen Fraud Shield is an AI-powered platform that detects phishing messages, digital arrest scams, fraudulent payment requests, and suspicious communications using NLP and machine learning.

Problem Statement

Cyber fraud, phishing attacks, fake banking notifications, and digital arrest scams are increasing rapidly. Existing systems often react after losses occur rather than preventing them.

Proposed Solution

The platform analyzes messages, emails, URLs, and payment requests, generates fraud risk scores, and provides actionable recommendations.

Objectives

Detect fraud in real time, reduce financial losses, improve public awareness, support reporting, and strengthen digital trust.

System Architecture

User Interface → Data Processing → AI Detection Layer → Risk Assessment Engine → Recommendation Engine → Reporting Module.

Technology Stack

Python, Streamlit, SQLite, Scikit-Learn, Transformers, Pandas, NumPy, GitHub.

Workflow

User submits suspicious content → preprocessing → AI analysis → risk score generation → recommendations → reporting options.

Key Features

Scam detection, fraud risk scoring, multilingual support, AI chatbot, reporting system, dashboard analytics.

Expected Benefits

Improved citizen safety, reduced fraud losses, enhanced reporting and law-enforcement support.

Innovation

Focuses on prevention through AI-powered real-time fraud detection and recommendations.

Scalability

Can be integrated with banking apps, telecom services, government portals, and cybercrime systems.

Future Enhancements

Voice scam detection, deepfake voice analysis, real-time call monitoring, fraud network visualization.

Conclusion

Citizen Fraud Shield helps create a safer digital ecosystem through proactive fraud detection and prevention.